A note on Open AI's "A counterexample to Hadwiger's conjecture"

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October 10, 2026

1 Summary

In the note they construct a triangle-free graph G that has many 4-cycles, such that for the complement graph we has its independence number $\alpha(\bar{G}) = 2$ and connected matching number $\mathrm{cm}(\bar{G}) < m/100$ where m is number of vertices. The connected matching number is defined as follows:

Definition 1. *A matching M in a graph H is called a* connected matching *if every two distinct edges of M are joined by an edge of H: for all distinct $e, f \in M$, there exist $u \in V(e)$ and $v \in V(f)$ such that $uv \in E(H)$. The* connected matching number *is*

$$\mathrm{cm}(H) = \max\{|M| : M \text{ is a connected matching in } H\}.$$

The complement graph provides a counterexample to Hadwiger's conjecture, which asserts that $\chi(H) \leq h(H)$, where $h(H)$ is the largest integer t such that H contains a K_t minor. Indeed, Proposition 3.5 of the paper (proved in Appendix A) gives

$$h(H) \leq \frac{|V(H)| + 4\,\mathrm{cm}(H) + 2}{3}.$$

Applying this to $H = \bar{G}$, and using $\alpha(\bar{G}) \leq 2$ so that every color class has at most two vertices, we obtain, for m large enough,

$$h(\bar{G}) \leq \frac{m + 4\,\mathrm{cm}(\bar{G}) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \leq \chi(\bar{G}).$$

Thus $h(\bar{G}) < \chi(\bar{G})$, contrary to Hadwiger's conjecture.

2 The construction

We retain the notation of the summary: G is the graph we constructed, and $H = \bar{G}$ is the proposed counterexample. All vector spaces and matrix operations below are over $\mathbb{F}_2$.

2.1 Fixed coefficient spaces

Fix the parameters g, b, j_*, h, n, N as in Section 2 of the manuscript, with g odd and $N = M_0 n$. Let

$$I = \mathbb{Z}/g\mathbb{Z}, \qquad \mathcal{E} = \binom{I}{2}, \qquad I_d = \{d+1, \dots, d+(g-1)/2\} \subset I.$$

The elements of I are tags and those of $\mathcal{E}$ are components. A Boolean point has a selector $s \in \mathbb{F}_2^b$ and base coordinates z in g^2+3 blocks of n bits, named $O_{d,t}$ $(d, t \in I)$, S, $\#$, and Z. Put

$$p_* = \sum_{j=0}^{j_*} \binom{b}{j}, \qquad B = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+(g^2+3)n}.$$

If p_s is the vector of evaluations of all squarefree selector monomials of degree at most j_*, define

$$v(s,z) = p_s \otimes (1,z), \qquad w(s,z) = v(s,z)v(s,z)^T.$$

For each tag ℓ, let W_ℓ be the span of these point moments subject to $O_{d,t} = 0$ whenever $\ell \notin I_d$. Define the cut-profile space

$$X = \left\{ (x_{\{\ell,\ell'\}})_{\{\ell,\ell'\} \in \mathcal{E}} : x_{\{\ell,\ell'\}} = w_\ell + w_{\ell'}, \quad w_\ell \in W_\ell \right\}.$$

Thus each component x_e is a matrix on B. The manuscript fixes a linear functional $a \in X^*$, a symmetric bilinear form T on X, and a prescribed self-frame matrix E on B (Equations (2.7)–(2.12)). Their detailed tester and mixing formulas are additional fixed ingredients; none is sampled separately for a graph vertex. Write $d_B = \dim B$ and $r(\lambda) = a + T(\lambda, \cdot)$.

2.2 The random variable attached to a vertex

For each component $e \in \mathcal{E}$, take common ambient spaces $V_e^+ = V_e^- = \mathbb{F}_2^N$, paired by the dot product. A raw vertex o is the tuple

$$o = (P_{o,e}, Q_{o,e}, X_{o,e}, Y_{o,e})_{e \in \mathcal{E}},$$

where $P_{o,e}, Q_{o,e}$ are $N \times d_B$ matrices and $X_{o,e}, Y_{o,e}$ are $N \times h$ matrices. For each component, sample uniformly among the quadruples satisfying

$$\begin{aligned} &[P_{o,e}\ X_{o,e}] \text{ and } [Q_{o,e}\ Y_{o,e}] \text{ have full column rank,} \\ &P_{o,e}^T Q_{o,e} = E, \\ &X_{o,e}^T Q_{o,e} = 0, \qquad P_{o,e}^T Y_{o,e} = 0. \end{aligned}$$

There is no prescribed value of $X_{o,e}^T Y_{o,e}$. Components are sampled independently; the four matrices within a component are constrained and hence need not be independent. Denote this finite raw vertex space by Ω_n and its probability law by μ_n. The frame matrices $X_{o,e}$ should not be confused with the witness space X.

Set $V = \bigoplus_{e\in\mathcal{E}} V_e^+ \otimes V_e^-$. Identifying tensors with matrices, a raw vertex determines

$$U_o(x) = (P_{o,e} x_e Q_{o,e}^T)_{e\in\mathcal{E}}, \qquad u_o(\xi) = \sum_{e\in\mathcal{E}} \langle Y_{o,e} X_{o,e}^T, \xi_e \rangle,$$

where $\langle A, B\rangle = \mathrm{Tr}(A^T B)$ is the entrywise pairing. The rank conditions make U_o injective.

2.3 Sampling the graph and declaring holes

Sample m independent raw vertices $o_1, \dots, o_m$ with law μ_n, where m is the sample size prescribed in Equation (2.4) of the manuscript. The graph vertices are the positions $1, \dots, m$, so repeated raw vertices are allowed.

For distinct positions i, j, declare a hole if there exist witnesses $\lambda_i, \lambda_j \in X$ satisfying

$$\begin{aligned} U_{o_i}\lambda_i &= U_{o_j}\lambda_j, \\ u_{o_j} \circ U_{o_i} &= r(\lambda_i), \qquad u_{o_i} \circ U_{o_j} = r(\lambda_j), \\ a(\lambda_i) + a(\lambda_j) &= 1. \end{aligned}$$

The middle equations are equalities of linear functionals on all of X. Put $ij \in E(G)$ precisely when there is a hole, and $ij \in E(H)$ otherwise. The hole relation is symmetric, loopless, and triangle-free by Lemma 2.2 of the manuscript. Consequently G is triangle-free and $\alpha(H) \le 2$, even if sampled raw vertices repeat.

Remark 1. *The initial random variable is the tuple of frame matrices o, not a Boolean point (s, z). Once $o_1, \dots, o_m$ are sampled, every graph edge is determined by the witness criterion. Boolean points vary when constructing witnesses from point moments; sampling such point parameters with the frames fixed is called a query in the manuscript.*

A The clique-minor bound (Proposition 3.5)

Proposition 1. *For every finite nonempty graph H with m vertices,*

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3}.$$

Proof. Write $c = \mathrm{cm}(H)$, $b = h(H)$. Thus that there are b pairwise disjoint, nonempty connected vertex sets, with an edge joining every pair of sets. These sets are the branch sets of a K_b minor. Let s be the number of sets with one vertex and e be the number of sets with two vertices.

Firstly $c > e + \frac{s-1}{2}$: Each two-vertex branch set has an edge. And pairs of singleton branches supports an edge, the set of them is a connected matching, so we have the desired inequality.

Then every remaining branch set has at least three vertices. Since the branch sets are disjoint,

$$m \geq s + 2e + 3(b - s - e) = 3b - 2s - e.$$

Combining two inequality give use the contradiction. □